

CompTIA.

A+ Core 2

Exam 220-1202

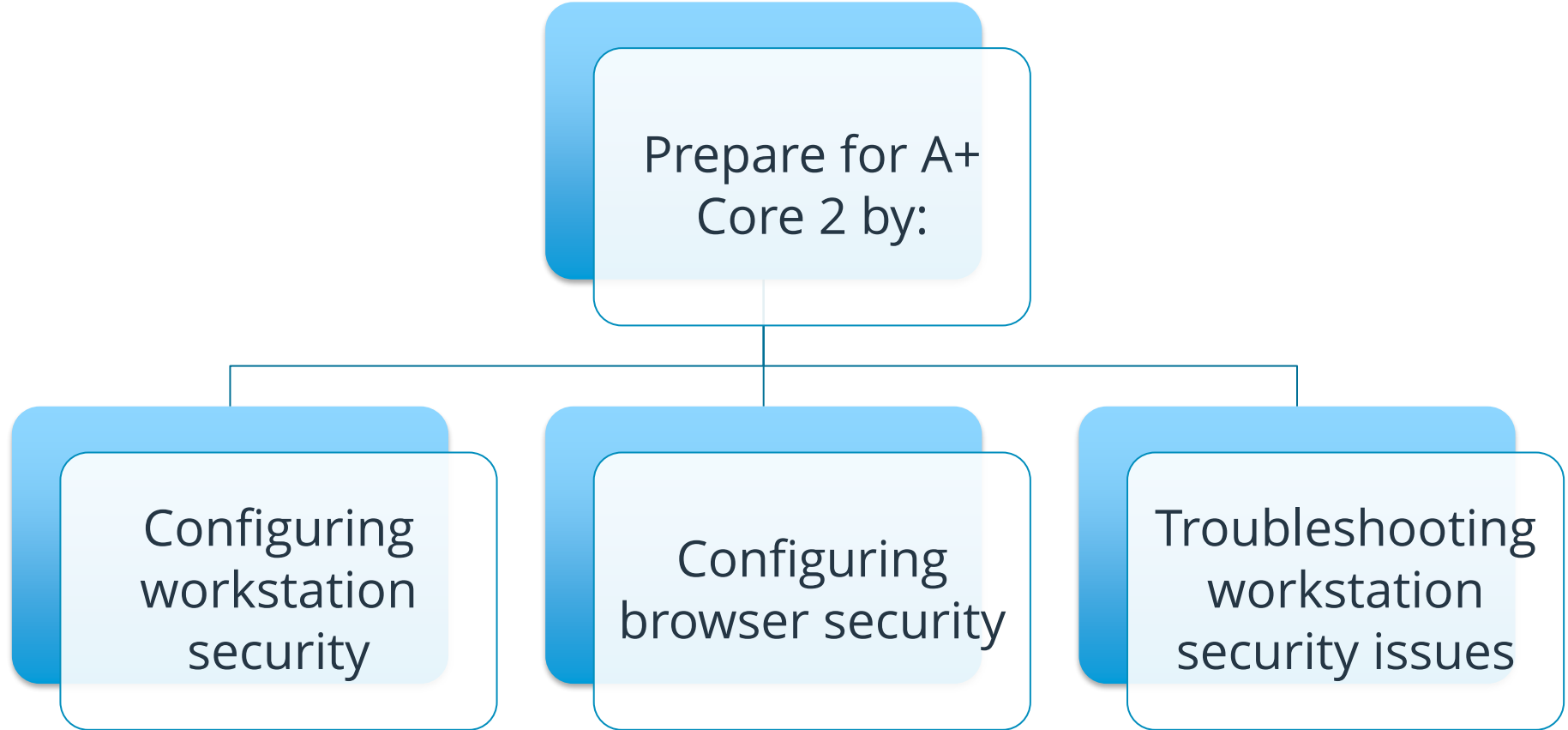
A+ 220-1202

Module 19



Managing Security Settings

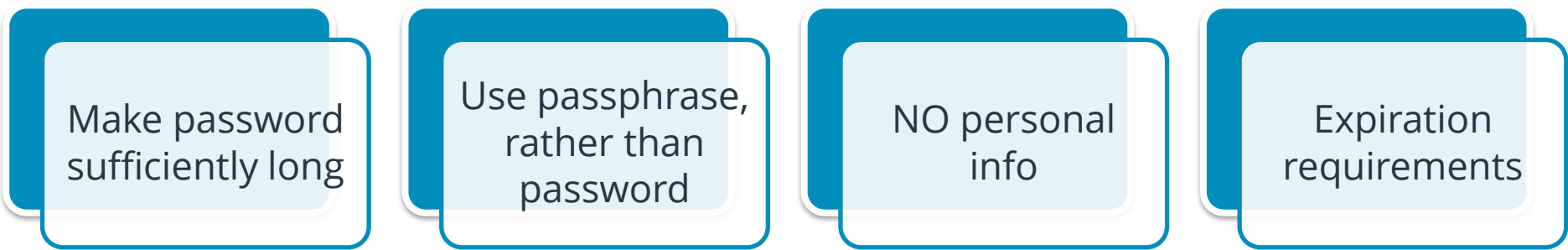
Learning Outcomes



Lesson 19.1

Account Security

Password Best Practices (Slide 1 of 2)

The slide features four blue rectangular boxes with rounded corners, each containing a password best practice. The boxes are arranged horizontally and have a slight overlap. Each box has a darker blue header area at the top and a lighter blue body area where the text is located. The text is in a dark blue, sans-serif font.

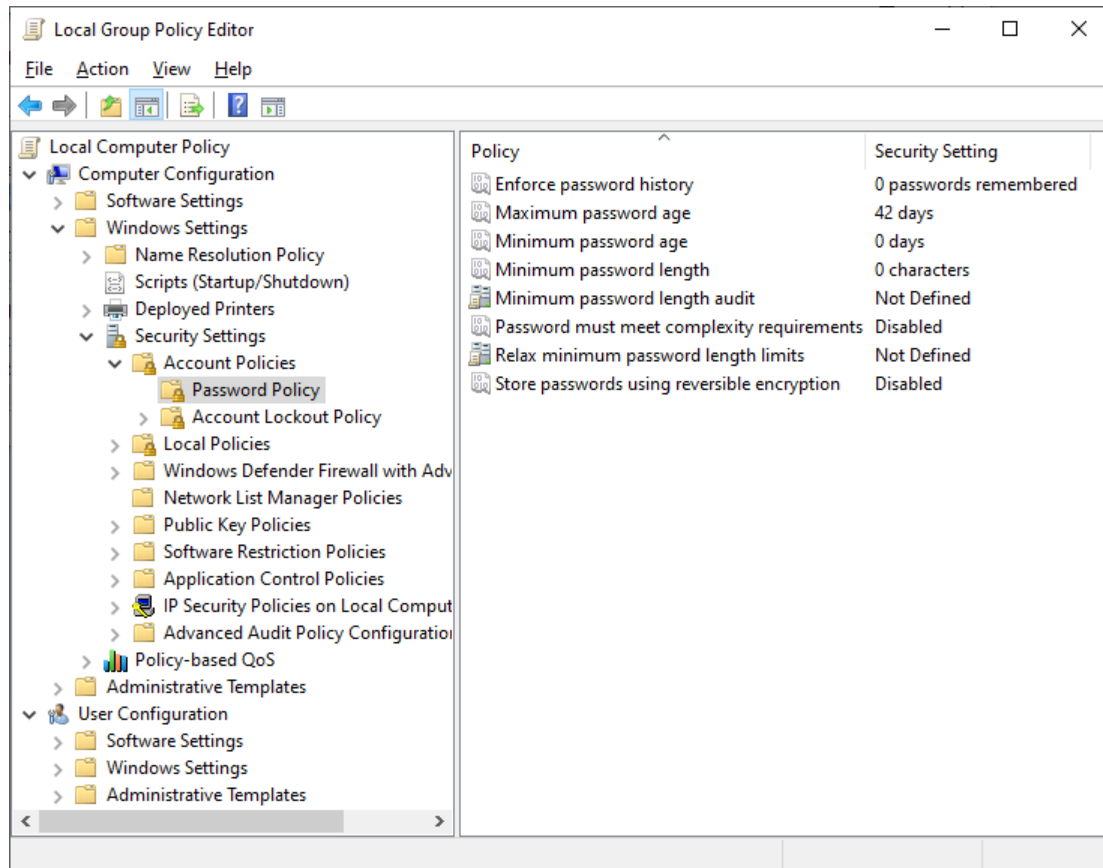
Make password
sufficiently long

Use passphrase,
rather than
password

NO personal
info

Expiration
requirements

Password Best Practices (Slide 2 of 2)



Screenshot courtesy of Microsoft.

End User Best Practices

01

Log off when
not using
system

- START+ L keys

02

Secure critical
systems from
theft

03

Secure PII and
passwords

04

Use password
manager

Restrict User Permissions

Least privilege principle

Fewest number of accounts with administrative rights as possible

Ensure UAC or sudo features are enabled

Change Default Administrator Account and Password

"Administrator" in WINDOWS and root in Linux/Unix; accounts should be changed and disabled from use

If not possible, change the default password



Create unique accounts with administrative access for admin use



Separation of duties

Disable Guest Account

Guest accounts do not require authentication



Disable all guest account use

Accountability for all users with unique accounts for each user

Account Policies (Slide 1 of 2)

- Restrict login times
 - Working hours/logon hours
- Account expiration
- Failed login attempt logout
- Concurrent logins
- Use timeout/screen locks

Account Policies (Slide 2 of 2)

David Martin Properties

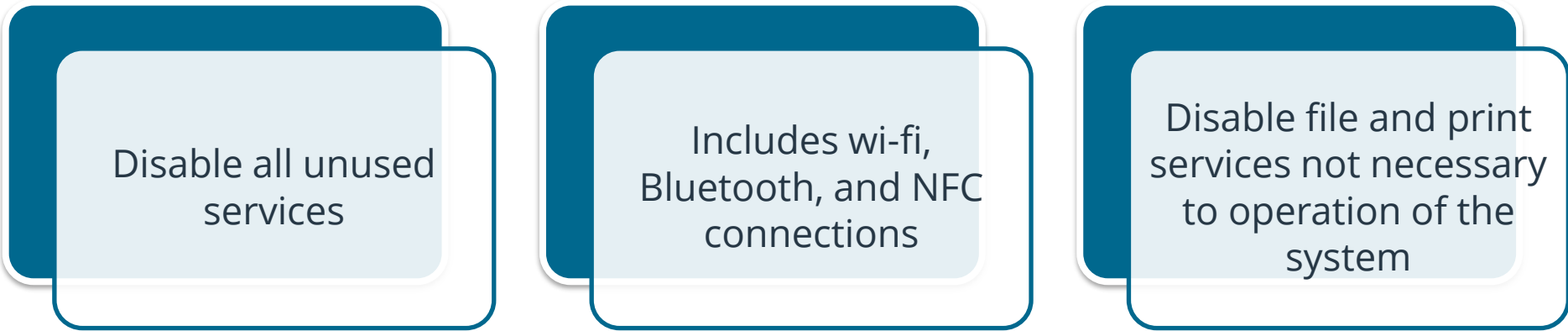
Member Of	Dial-in	Environment	Sessions
Remote control	Remote Desktop Services Profile	COM+	

General	Address	Account	Profile	Telephones	Organization
User logon name: dmartin @classroom.local					
User logon name (pre-Windows 2000): classroom\ dmartin					
Logon Hours... Log On To...					
☒ Unlock account					
Account options: ☒ User must change password at next logon ☐ User cannot change password ☐ Password never expires ☐ Store password using reversible encryption					
Account expires: ☒ Never ☐ End of: Tuesday , October 30, 2018					

OK Cancel Apply Help

Screenshot courtesy of Microsoft.

Unused Services



Disable all unused services

The diagram consists of three light blue rounded rectangular boxes with dark blue borders, arranged horizontally. Each box contains text describing a step in a process. The first box says 'Disable all unused services'. The second box says 'Includes wi-fi, Bluetooth, and NFC connections'. The third box says 'Disable file and print services not necessary to operation of the system'.

Includes wi-fi,
Bluetooth, and NFC
connections

Disable file and print
services not necessary
to operation of the
system

Lab Activity

- Lab: Enforce Password Settings
(available in CertMaster Learn and CertMaster Perform)
- Lab: User Policy Management
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Configure Domain Account Security
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Account Security

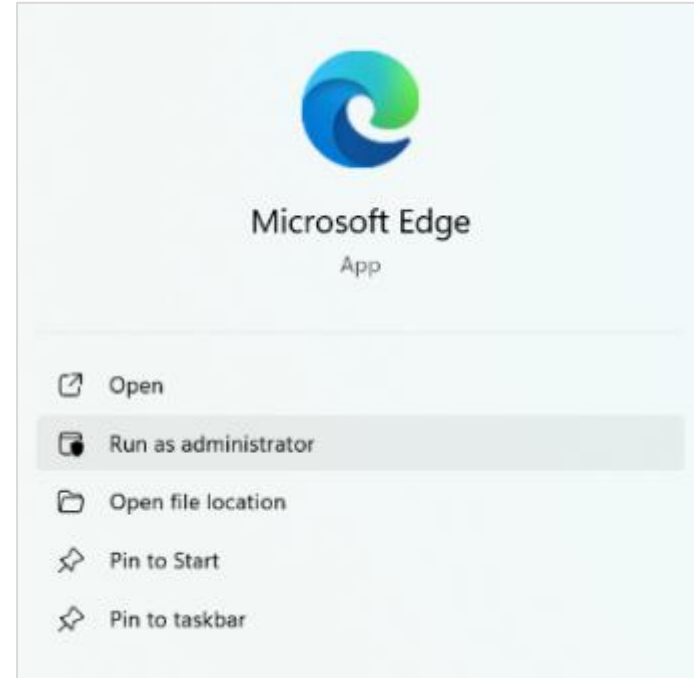
- Password Best Practices
- End User Best Practices
- Restrict User Permissions
- Change Default Administrator Account and Password
- Disable Guest Account
- Account Policies
- Unused Services

Lesson 19.2

Workstation Security

Execution Control

- Restrict use of elevated permissions
- "Run As administrator"
 - UAC warning
 - Admin account login requirement



Screenshot courtesy of Microsoft.

Trusted/Untrusted Software Sources

Download

Download from trusted sources



Scan

Scan all downloads



Verify

Verify digital signatures and/or file hash



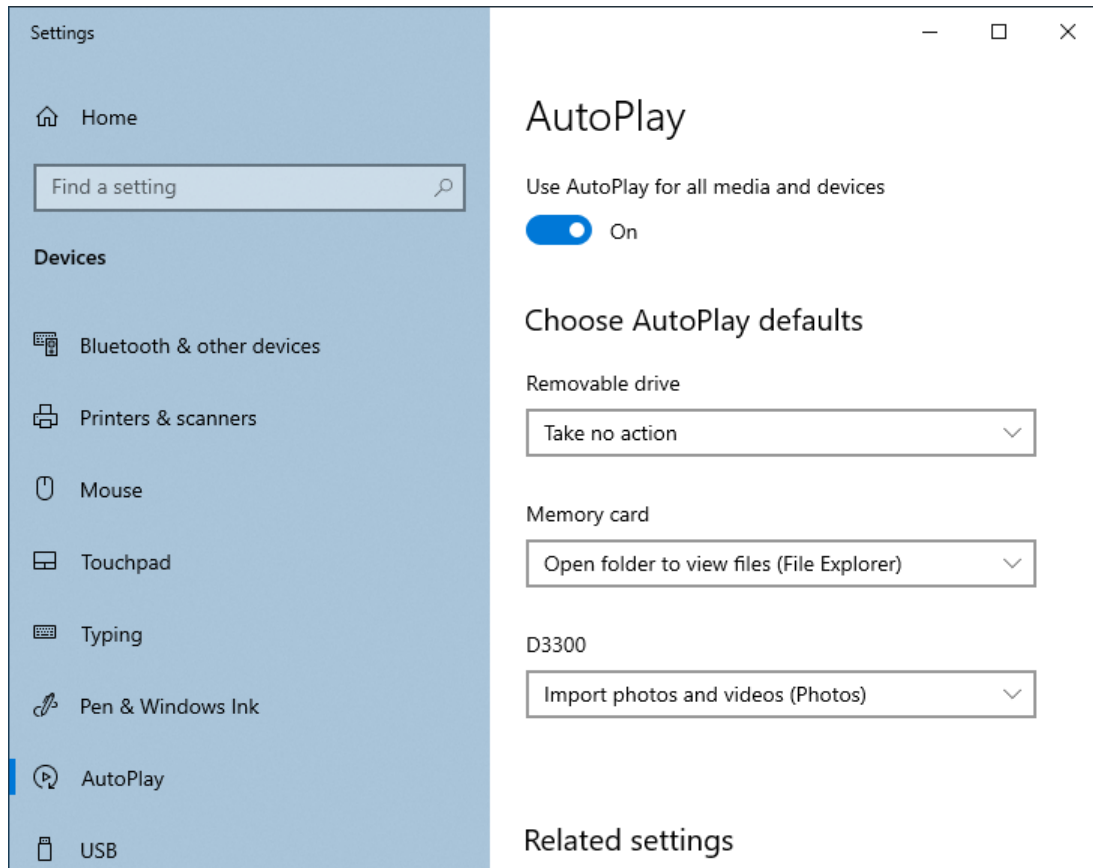
Remove

Remove all unnecessary software applications

AutoRun and AutoPlay (Slide 1 of 2)

- Disable Autorun and AutoPlay
 - Autorun.inf
- Configured under a drive's property dialog box

AutoRun and AutoPlay (slide 2 of 2)



Screenshot courtesy of Microsoft.

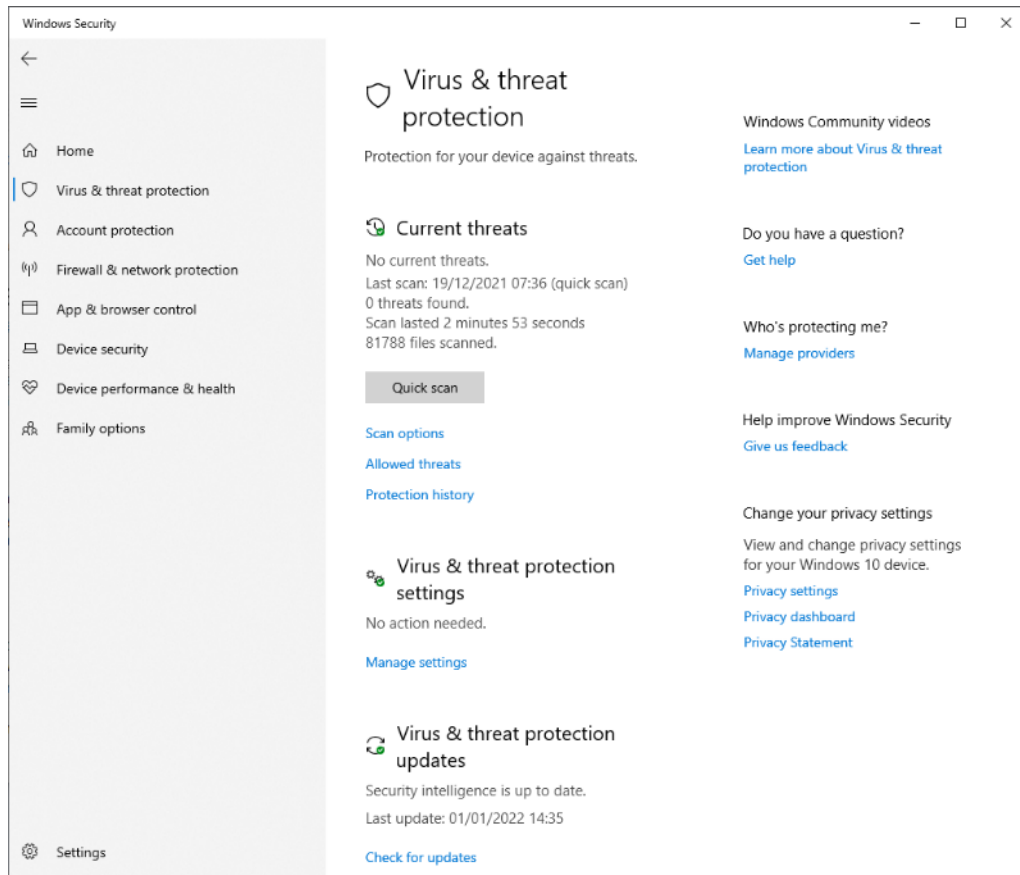
Windows Defender Antivirus (Slide 1 of 2)

Detect and prevent execution of malware

Host-based system

- Settings and rules are only for that single node, unless managed through group policy

Windows Defender Antivirus (slide 2 of 2)



Screenshot courtesy of Microsoft.

Windows Defender Antivirus Updated Definitions

Definitions

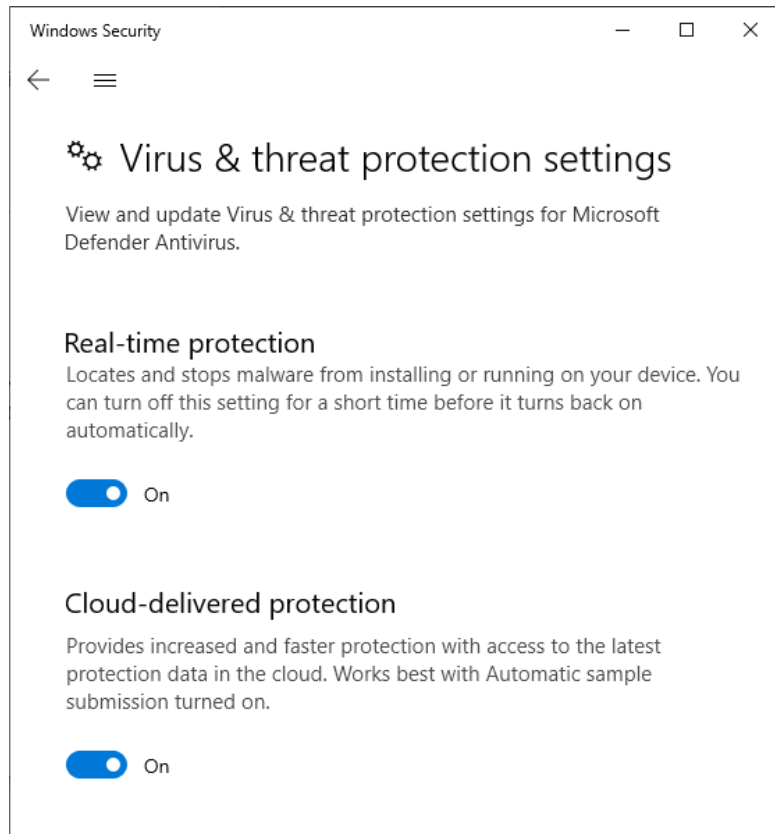
- Identify malware using signatures and heuristic models
- Daily and sometimes hourly updates

Scan engine

- Improves software performance

Activating and Deactivating Windows Defender Antivirus

- Temporary disable
 - Real-time protection slider
- Third-party antivirus/
antimalware software installed



Screenshot courtesy of Microsoft.

Windows Defender Firewall (Slide 1 of 3)



Inbound/outbound Rules



Port security

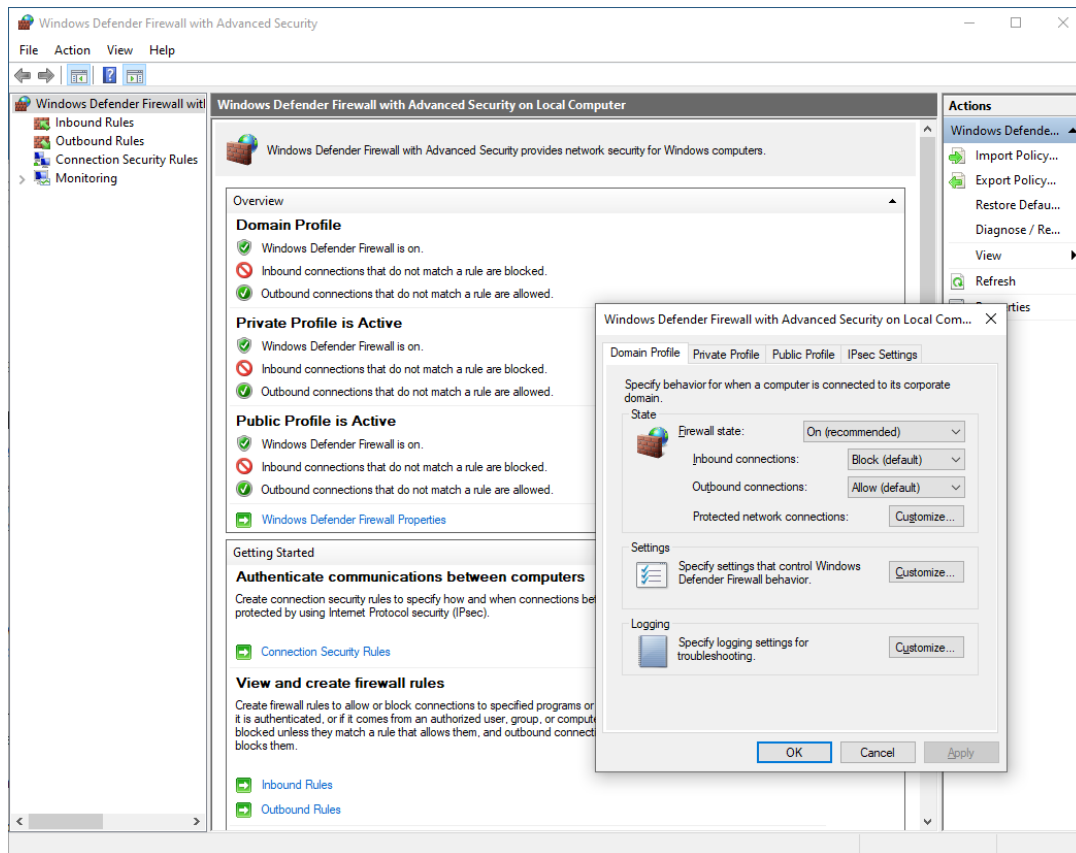


Application security



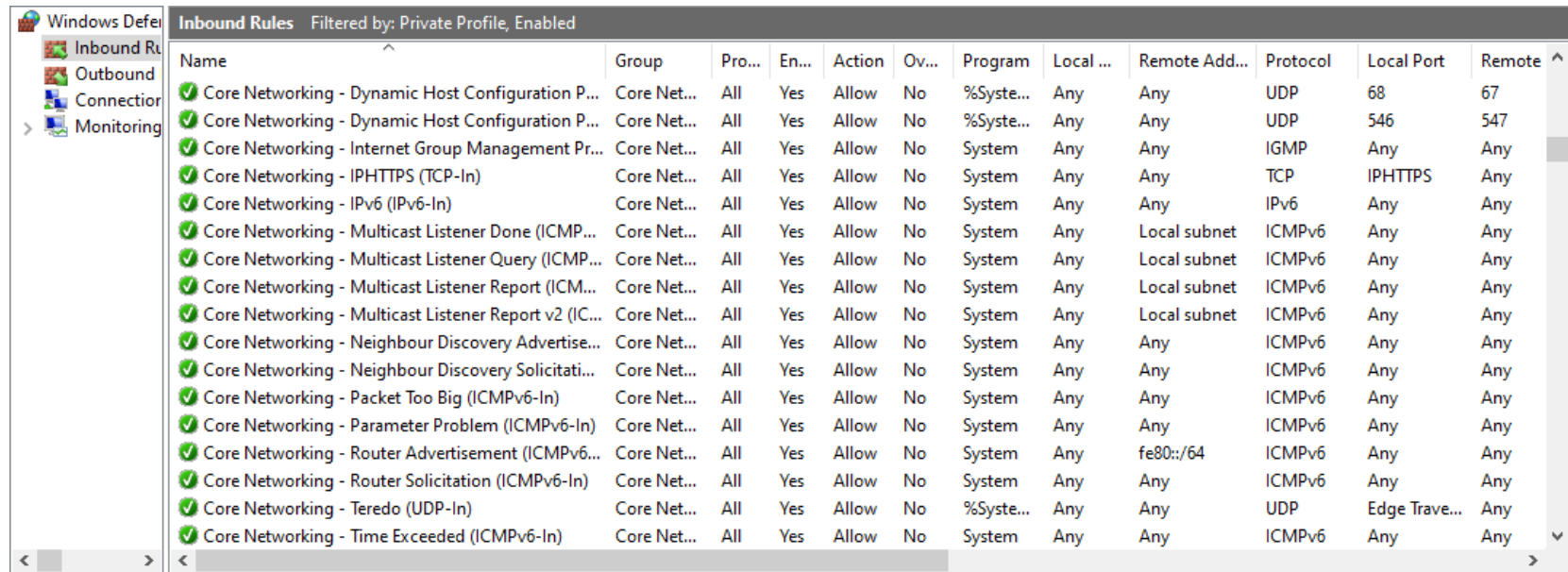
Triggers

Windows Defender Firewall (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Windows Defender Firewall (Slide 3 of 3)



The screenshot shows the Windows Defender Firewall 'Inbound Rules' window. The left sidebar contains icons for 'Inbound Rules', 'Outbound Rules', 'Connections', and 'Monitoring'. The main pane is titled 'Inbound Rules' and 'Filtered by: Private Profile, Enabled'. It displays a list of 17 inbound rules, all of which are enabled and have a 'Allow' action. The rules are grouped under 'Core Networking'. The columns in the table are: Name, Group, Protocol, Enabled, Action, Overlaid, Program, Local Address, Remote Address, Protocol, Local Port, and Remote Port.

Name	Group	Pro...	En...	Action	Ov...	Program	Local ...	Remote Add...	Protocol	Local Port	Remote
Core Networking - Dynamic Host Configuration P...	Core Net...	All	Yes	Allow	No	%Syste...	Any	Any	UDP	68	67
Core Networking - Dynamic Host Configuration P...	Core Net...	All	Yes	Allow	No	%Syste...	Any	Any	UDP	546	547
Core Networking - Internet Group Management Pr...	Core Net...	All	Yes	Allow	No	System	Any	Any	IGMP	Any	Any
Core Networking - IPHTTPS (TCP-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	TCP	IPHTTPS	Any
Core Networking - IPv6 (IPv6-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	IPv6	Any	Any
Core Networking - Multicast Listener Done (ICMP...	Core Net...	All	Yes	Allow	No	System	Any	Local subnet	ICMPv6	Any	Any
Core Networking - Multicast Listener Query (ICMP...	Core Net...	All	Yes	Allow	No	System	Any	Local subnet	ICMPv6	Any	Any
Core Networking - Multicast Listener Report (ICM...	Core Net...	All	Yes	Allow	No	System	Any	Local subnet	ICMPv6	Any	Any
Core Networking - Multicast Listener Report v2 (IC...	Core Net...	All	Yes	Allow	No	System	Any	Local subnet	ICMPv6	Any	Any
Core Networking - Neighbour Discovery Advertise...	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any
Core Networking - Neighbour Discovery Solicitati...	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any
Core Networking - Packet Too Big (ICMPv6-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any
Core Networking - Parameter Problem (ICMPv6-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any
Core Networking - Router Advertisement (ICMPv6...	Core Net...	All	Yes	Allow	No	System	Any	fe80::/64	ICMPv6	Any	Any
Core Networking - Router Solicitation (ICMPv6-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any
Core Networking - Teredo (UDP-In)	Core Net...	All	Yes	Allow	No	%Syste...	Any	Any	UDP	Edge Trave...	Any
Core Networking - Time Exceeded (ICMPv6-In)	Core Net...	All	Yes	Allow	No	System	Any	Any	ICMPv6	Any	Any

Screenshot courtesy of Microsoft.

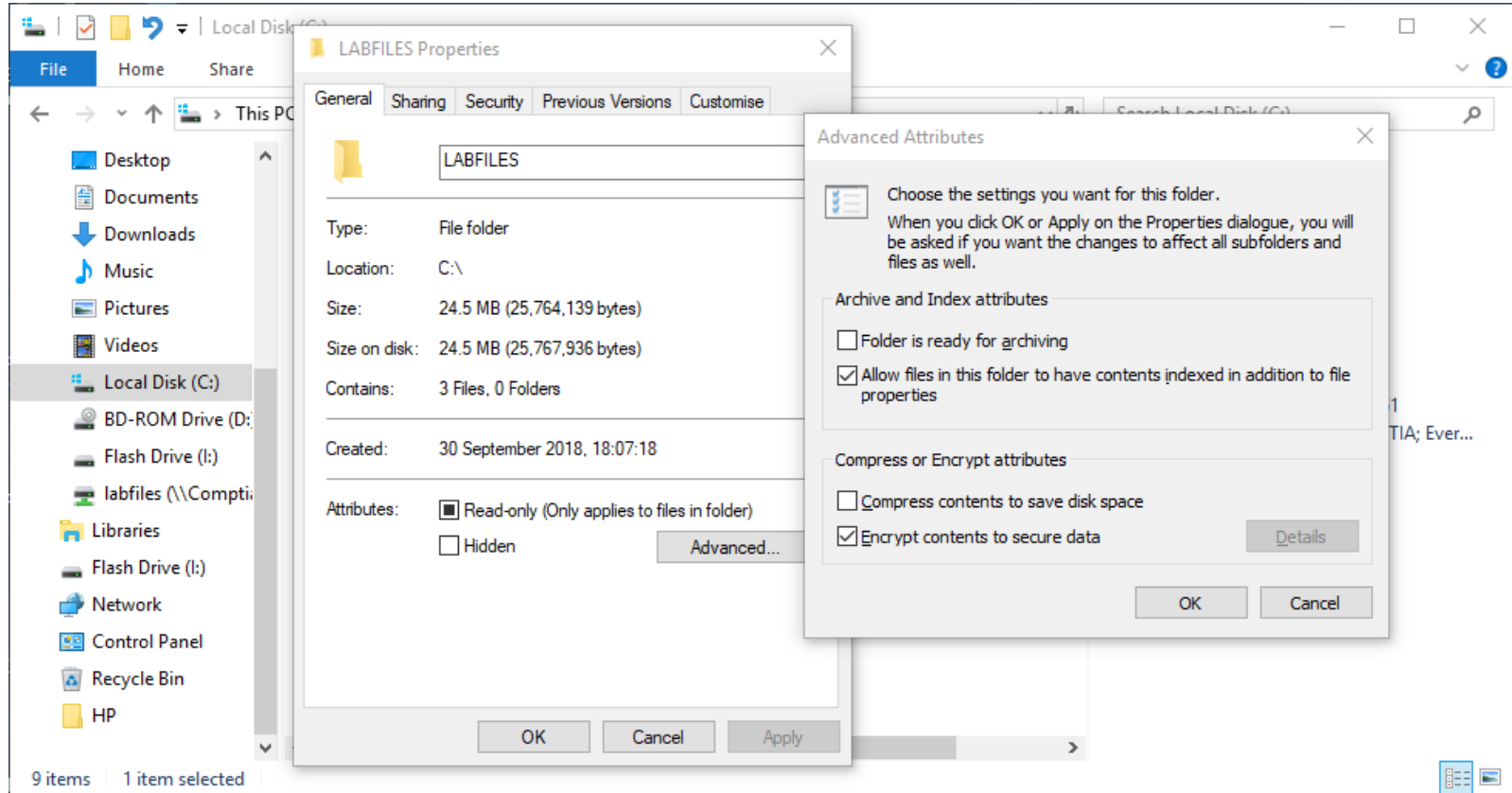
Encrypting File System (Slide 1 of 3)

Built into Windows
to protect data at
rest

- Not available on Home Edition

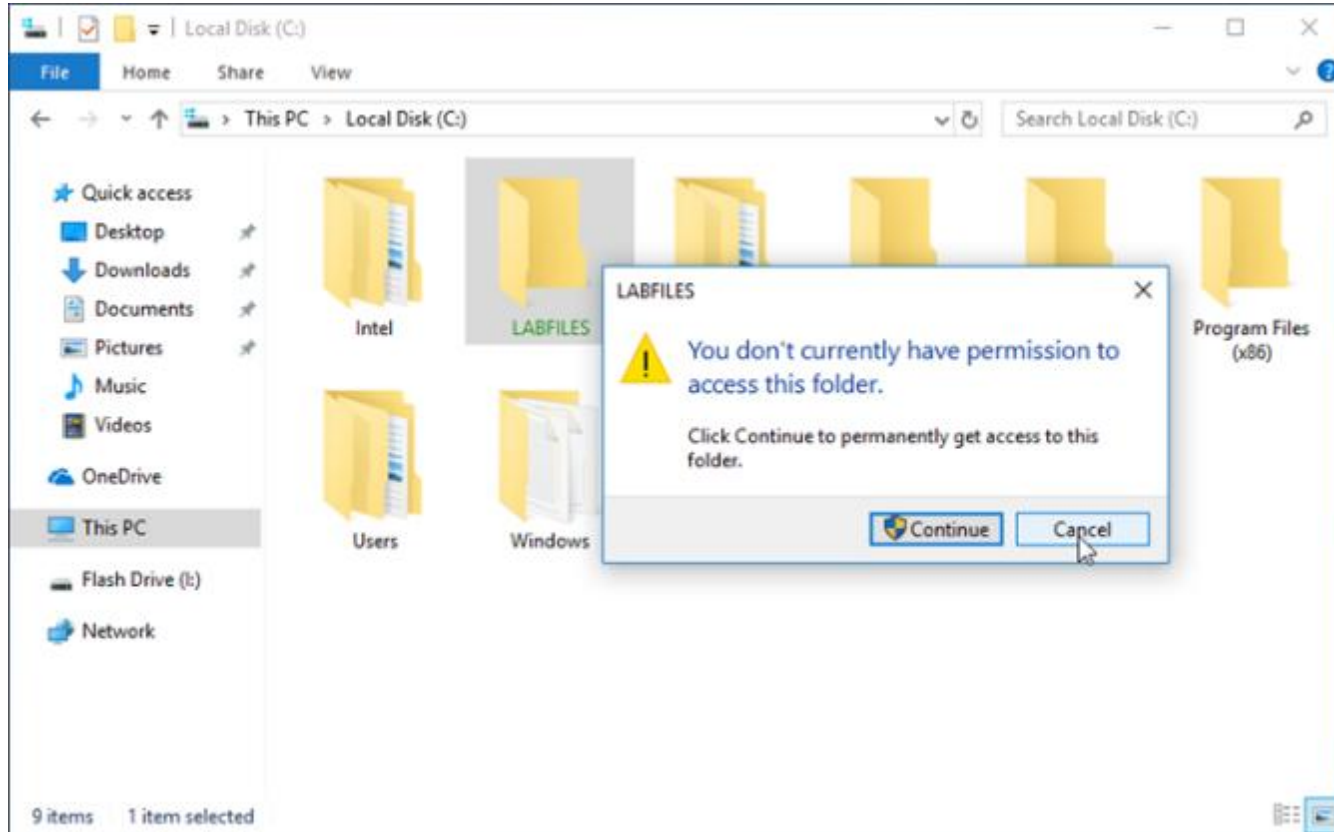
Advanced button of
file's property dialog
box

Encrypting File System (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Encrypting File System (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Windows BitLocker and BitLocker To Go (Slide 1 of 3)

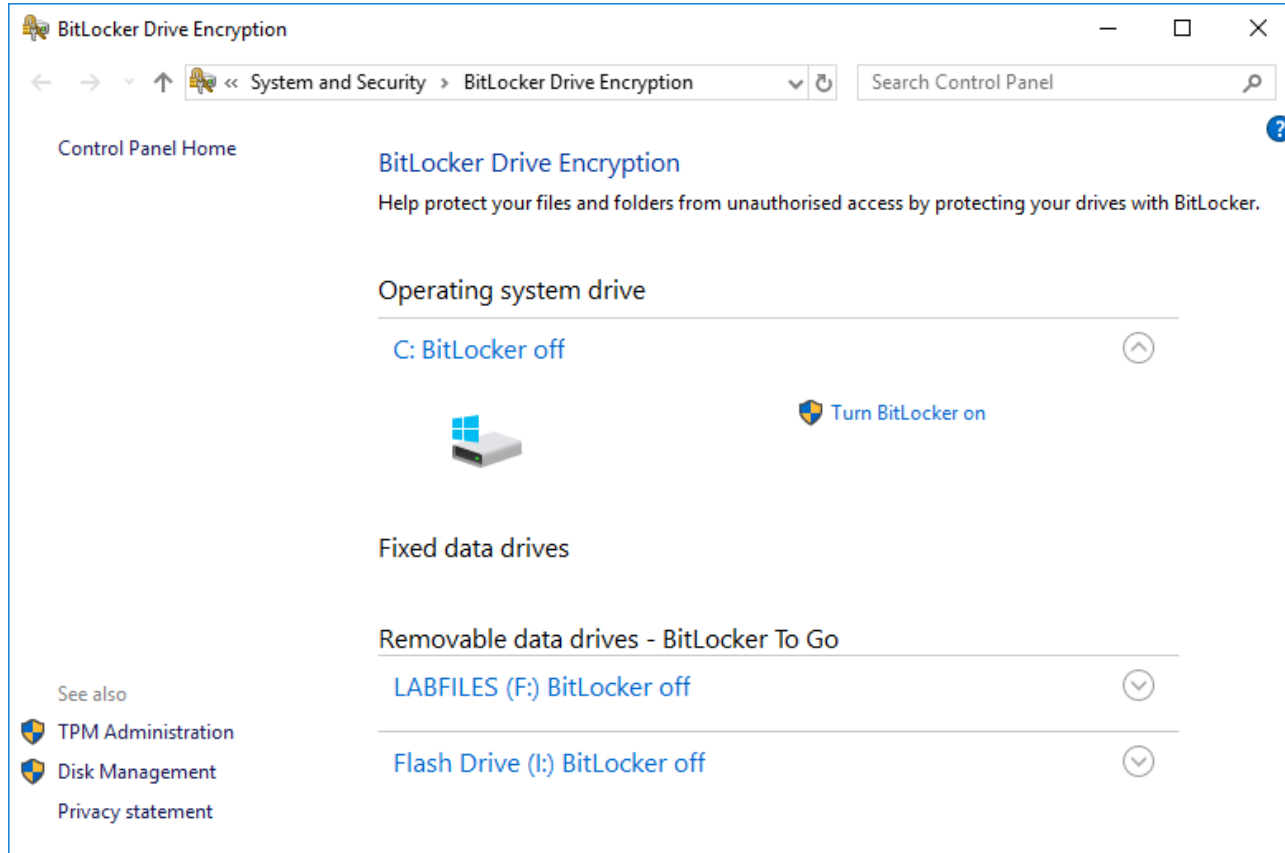
Included with all versions of Windows except Home edition

Full disk encryption software

- Password protected
- Recovery Key backup- safe location outside of the disk being protected
 - TPM can be used

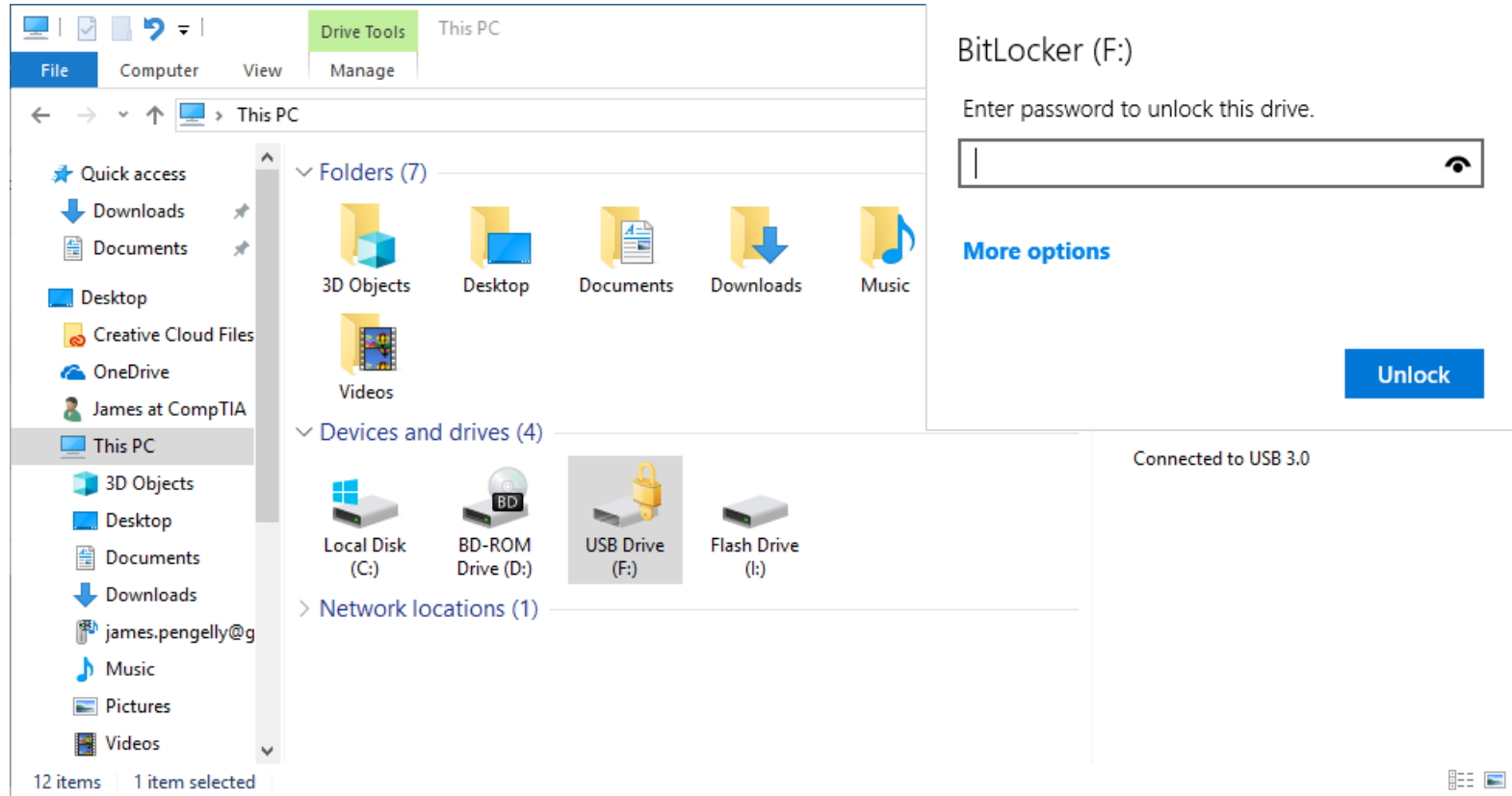
To Go- removable drive version

Windows BitLocker and BitLocker To Go (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Windows BitLocker and BitLocker To Go (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Lab Activity

- Lab: Configure Microsoft Defender Antivirus
(available in CertMaster Learn and CertMaster Perform)
- Lab: Encrypt Files with EFS
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure BitLocker with a TPM
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Analyze Workstation Security Settings
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Workstation Security

- Execution Control
- Trusted/Untrusted Software Sources
- AutoRun and AutoPlay
- Windows Defender Antivirus
- Windows Defender Antivirus Updated Definitions
- Activating and Deactivating Windows Defender Antivirus
- Windows Defender Firewall
- Encrypting File System
- Windows BitLocker and BitLocker To Go

Lesson 19.3

Browser Security

Browser Selection and Installation (Slide 1 of 2)

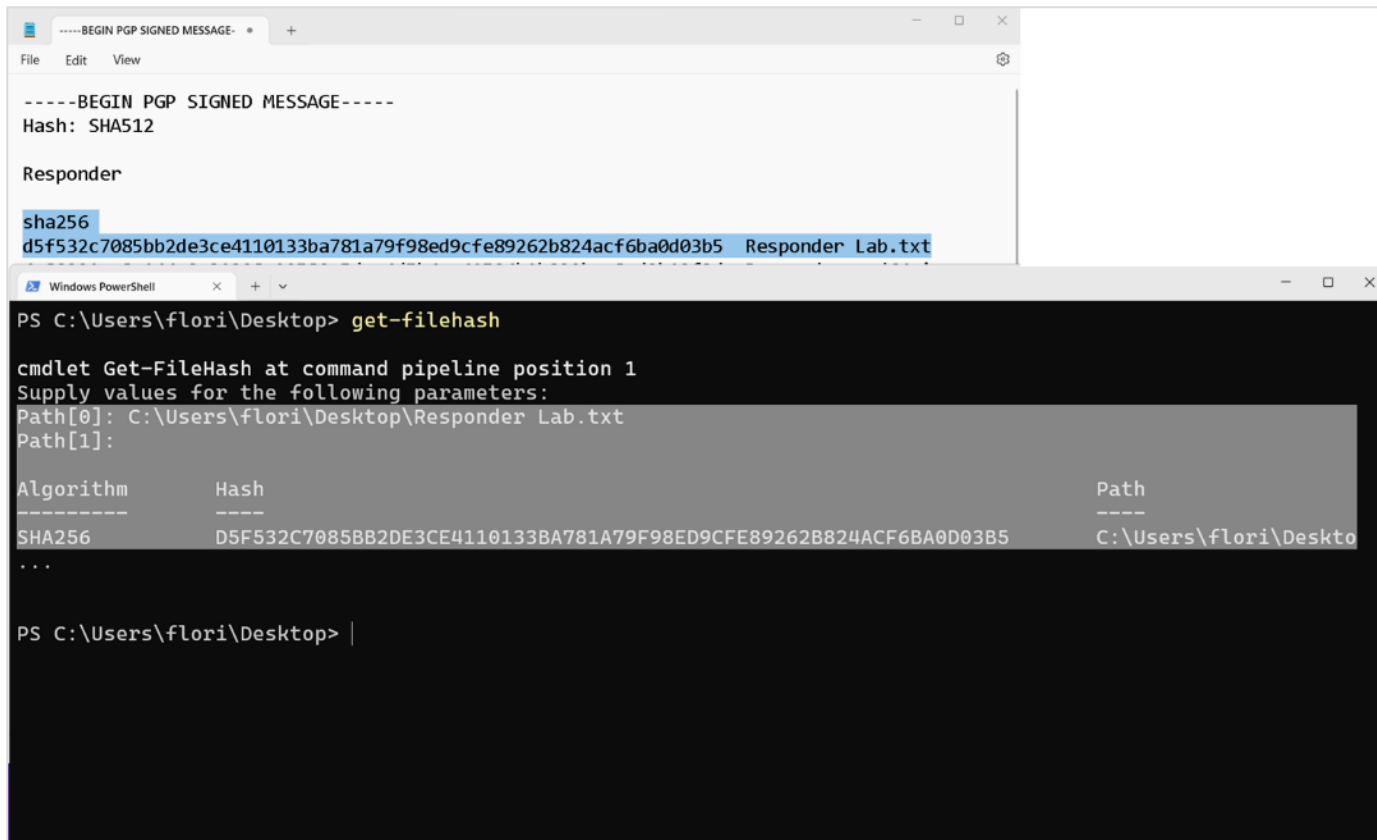
Trusted sources

- Verify download using hash or digital signature

Untrusted sources

- May expose system to malware including advertising spam
- Should not be used

Browser Selection and Installation (Slide 2 of 2)



-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA512

Responder

sha256
d5f532c7085bb2de3ce4110133ba781a79f98ed9cfe89262b824acf6ba0d03b5 Responder Lab.txt

Windows PowerShell

```
PS C:\Users\flori\Desktop> get-filehash
```

cmdlet Get-FileHash at command pipeline position 1
Supply values for the following parameters:
Path[0]: C:\Users\flori\Desktop\Responder Lab.txt
Path[1]:

Algorithm	Hash	Path
SHA256	D5F532C7085BB2DE3CE4110133BA781A79F98ED9CFE89262B824ACF6BA0D03B5	C:\Users\flori\Desktop
...		

```
PS C:\Users\flori\Desktop> |
```

Screenshot courtesy of Microsoft.

Browser Settings (Slide 1 of 2)



Sign-in and data sync

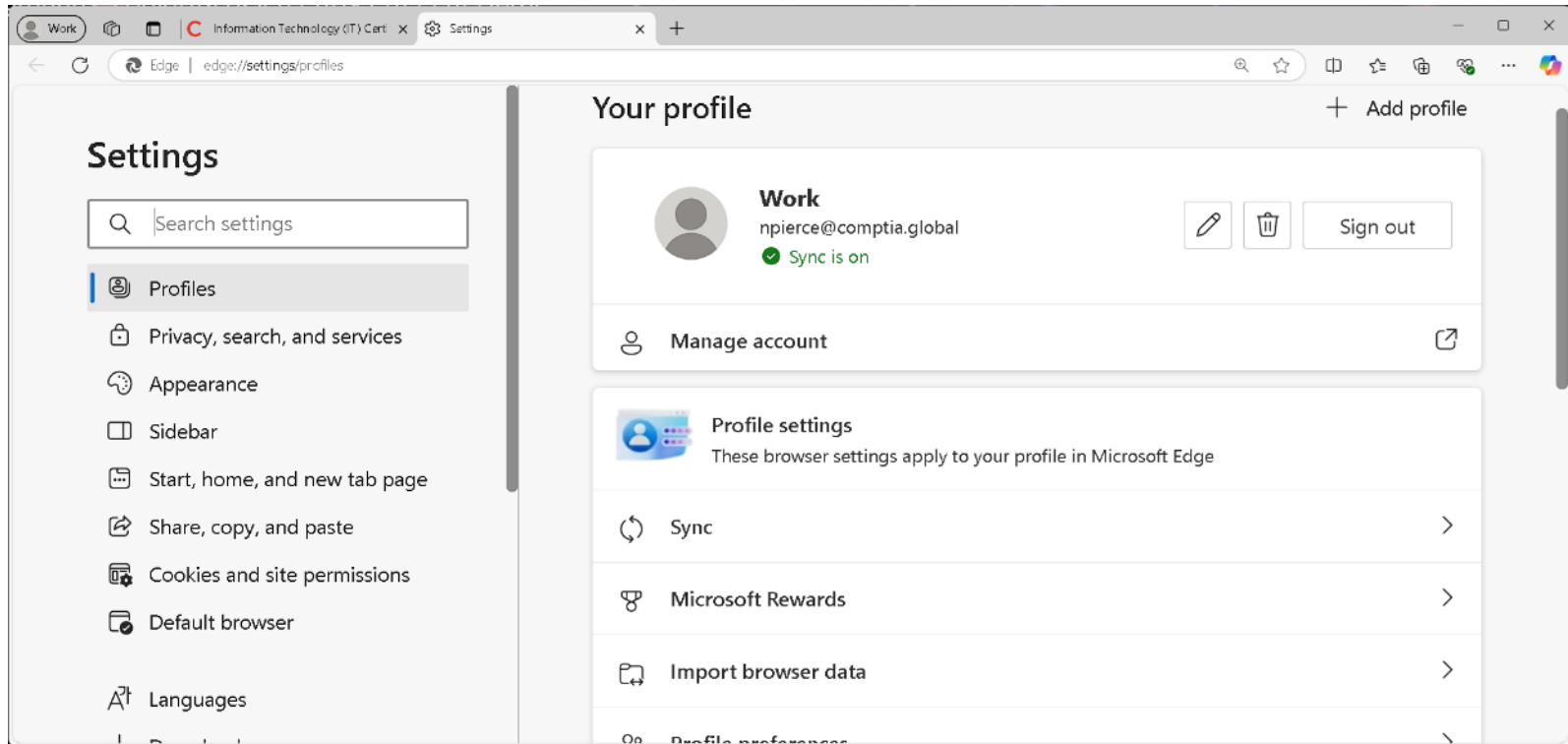
Cross platform and device sync



Password manager

Holds passwords securely and can sync across devices

Browser Settings (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Browser Extensions and Plug-ins (Slide 1 of 2)

Extension- Add or change a feature

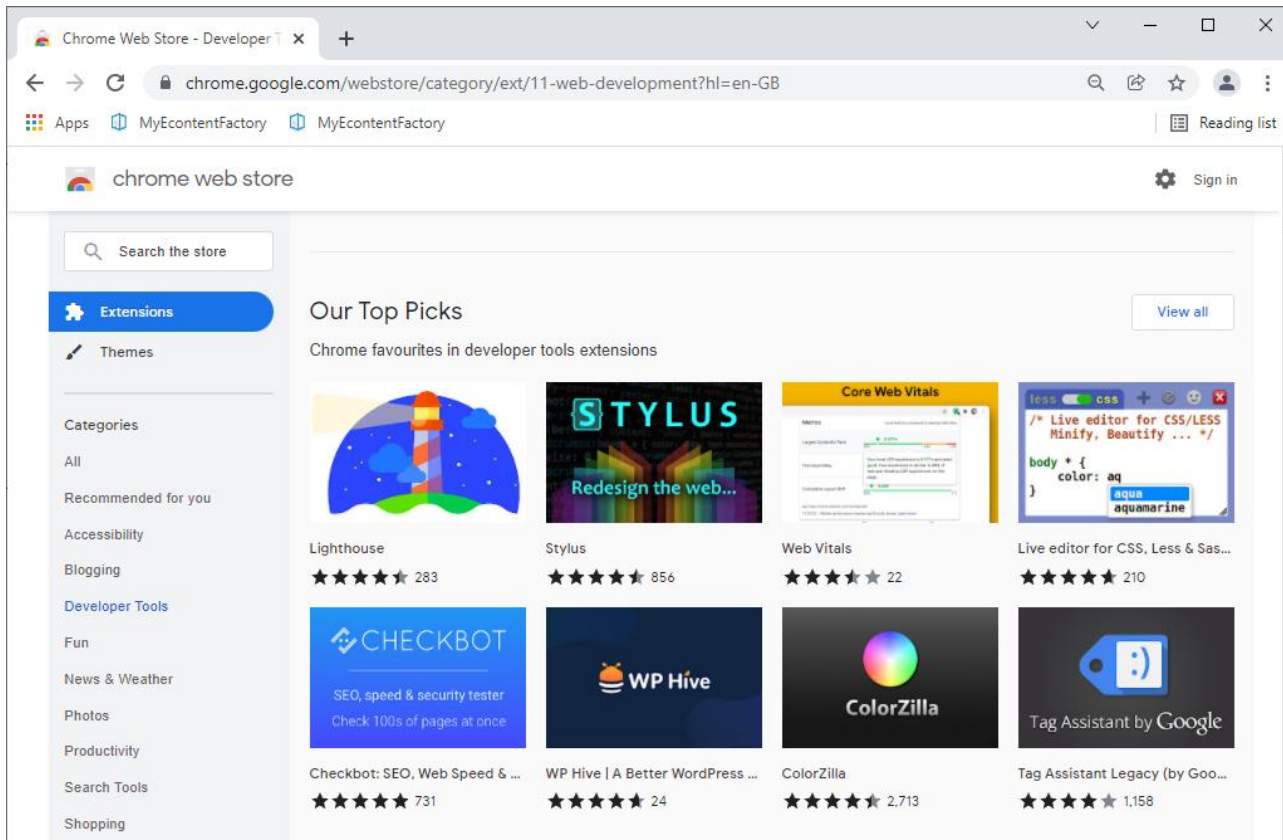
Plug-in- Support various types of content

Trusted sources

- Browser web store

Only enable when needed

Browser Extensions and Plug-ins (slide 2 of 2)



Screenshot courtesy of Google, a trademark of Google LLC.

Browser Patching

Updates- processed through OS updates or separately

Auto update upon launch

About menu

Secure Connections and Valid Certificates (Slide 1 of 3)



SSL/TLS certificate



Certificate Authority (CA)



Validity

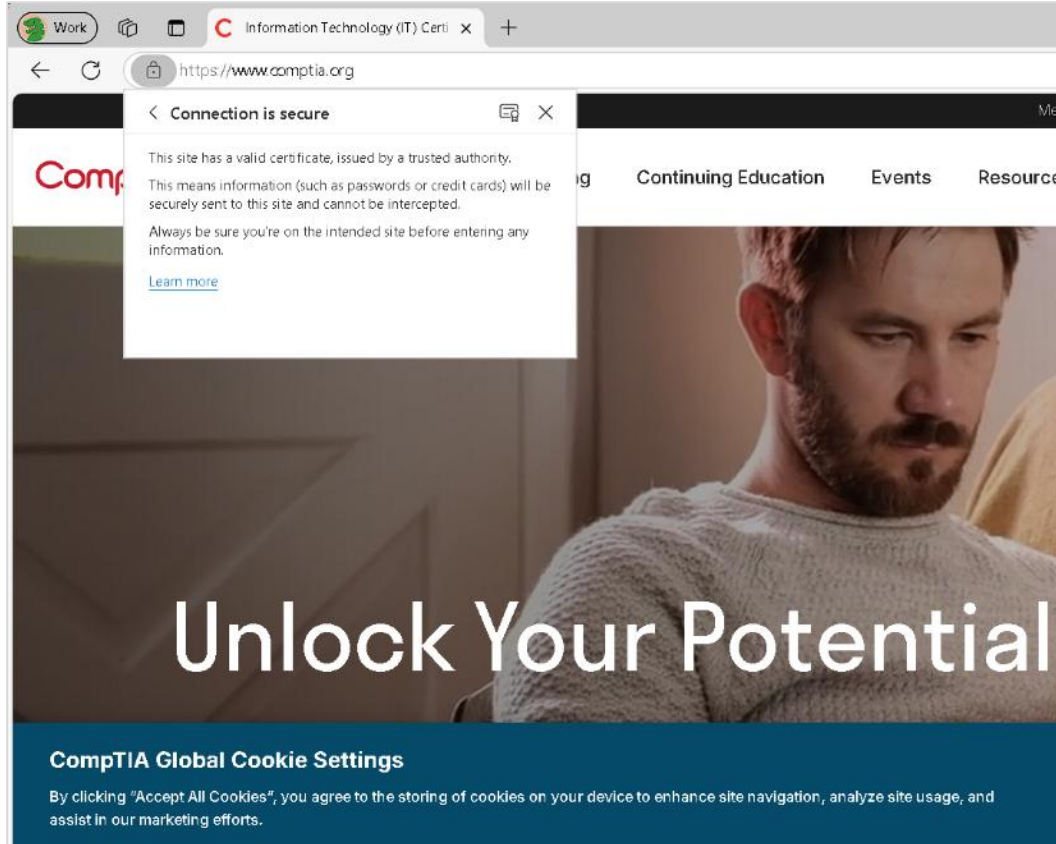


Asymmetric Algorithm

RSA

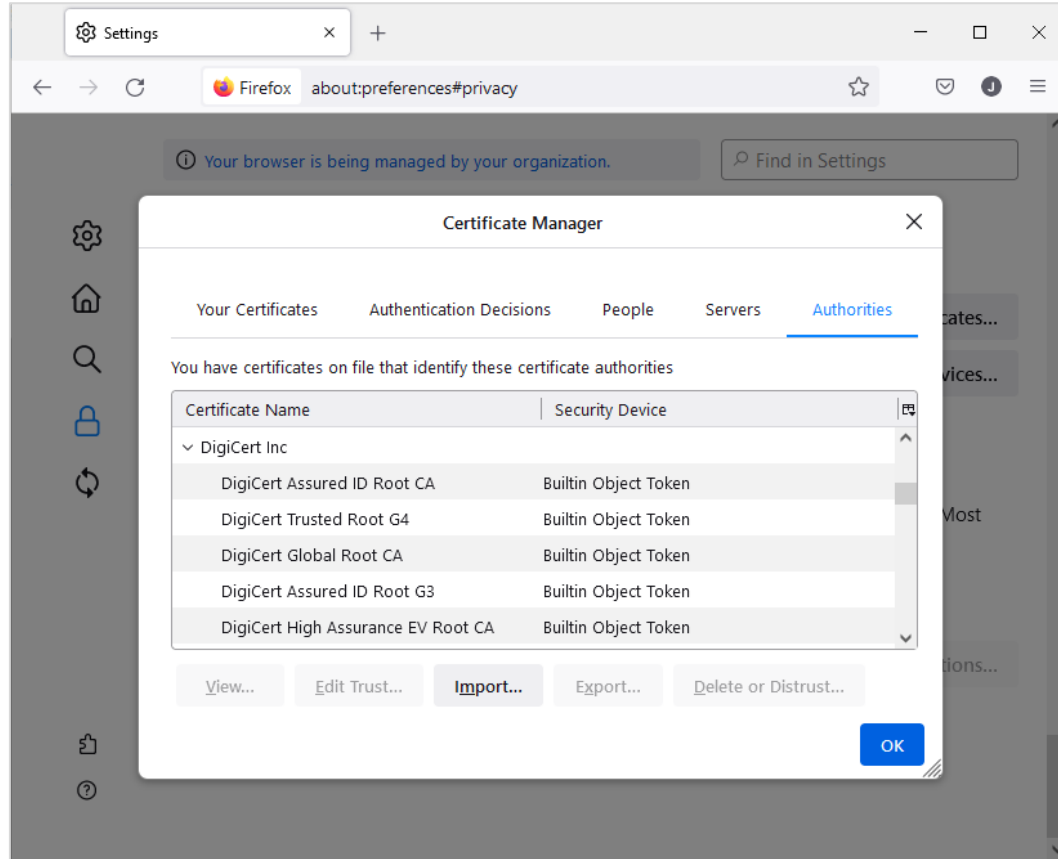
ECC

Secure Connections and Valid Certificates (Slide 2 of 3)



Screenshot courtesy of CompTIA and Edge.

Secure Connections and Valid Certificates (Slide 3 of 3)



Screenshot courtesy of Mozilla.

Browser Privacy Settings (Slide 1 of 2)



Cookies



Pop-up blockers



Ad blockers



Proxy settings



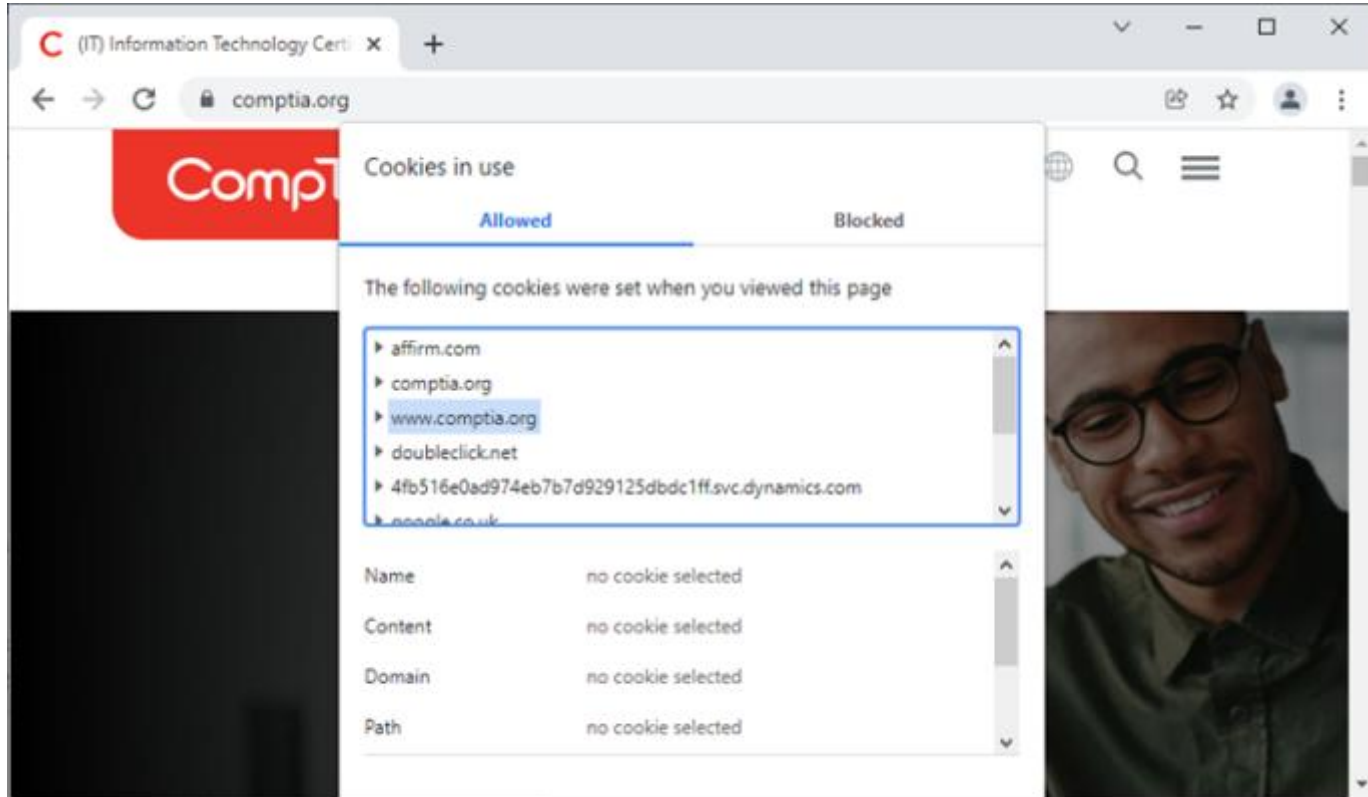
Clearing cache



Private browsing

Does not hide your web browsing from your ISP!

Browser Privacy Settings (Slide 2 of 2)



Screenshot courtesy of CompTIA and Google, a trademark of Google, LLC.

Lab Activity

- Lab: Practice Installing & Removing Browser Software
(available in CertMaster Learn and CertMaster Perform)
- Lab: Safe Browsing Best Practices
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Configure Browser Security
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Browser Security

- Browser Selection and Installation
- Browser Settings
- Browser Extensions and Plug-ins
- Browser Patching
- Secure Connections and Valid Certificates
- Browser Privacy Settings

Lesson 19.4

Troubleshoot Workstation Security

Malware Vectors



Virus



Boot sector virus



Trojan



Worm



Fileless malware

Lives in memory, not
within a file

Malware Payloads (Slide 1 of 2)

- Actions does the malware accomplish
- Backdoor
 - Ports, services, accounts
- Adware
- Spyware and keyloggers
- Rootkits

Malware Payloads (Slide 2 of 2)

```
meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > keyscan_start
Starting the keystroke sniffer ...
meterpreter > keyscan_dump
Dumping captured keystrokes ...
https<Right Shift>://tickets.structureality.com/scp<CR>
jaime<Tab><Right Shift>Pa<Right Shift>$$w0rd
meterpreter > █
```

Ransomware and Cryptominers (Slide 1 of 2)



Hold data "hostage" until ransom is paid



Mine cryptocurrency using host resources such as CPU and GPU

Ransomware and Cryptominers (Slide 2 of 2)



Screenshot courtesy of Wikimedia.

Troubleshoot PC Security Symptoms



Performance symptoms



Application crashes



Service issues



File system errors and anomalies

Desktop Alerts and Notifications

Push
notifications to
the OS

False or fake
alerts that
confuse user

Endpoint Monitoring Solutions



Monitor for changes on endpoint nodes



Provide early alerts that something may be wrong



EDR (Endpoint Detection & Response)

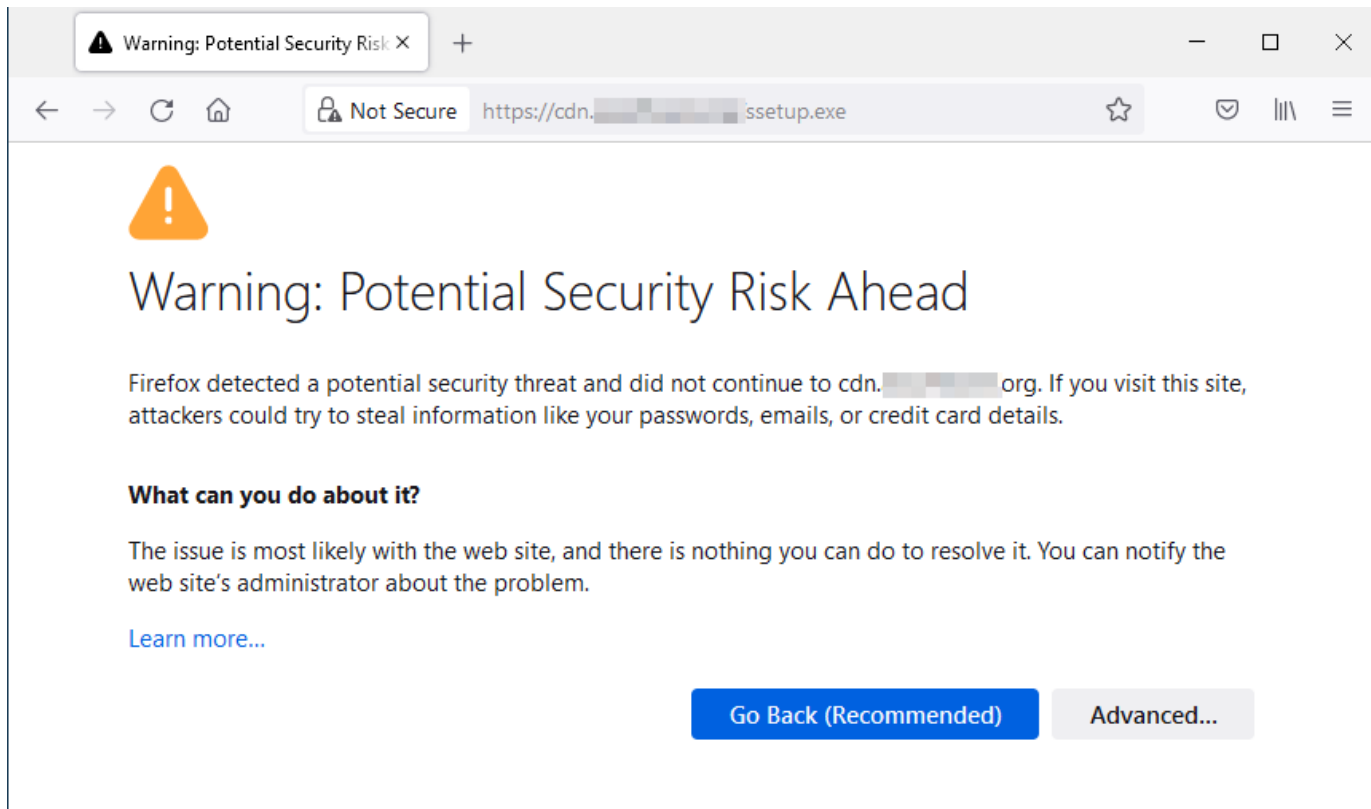
MDR (Managed Detection & Response)

XDR (Extended Detection & Response)

Troubleshoot Browser Symptoms (Slide 1 of 2)

- Redirection to another webpage or website
- Certificate warnings
 - Not necessarily an indication of malicious website, but should be looked at closely before proceeding

Troubleshoot Browser Symptoms (Slide 2 of 2)



Screenshot courtesy of Mozilla.

Best Practices for Malware Removal (Slide 1 of 4)

1. Investigate and verify malware symptoms.
2. Quarantine infected systems.
3. Disable System Restore in Windows.
4. Remediate infected systems:
 - Update anti-malware software.
 - Scanning and removal techniques (e.g., safe mode, Reimage/Reinstall environment).
5. Schedule scans and run updates.

Best Practices for Malware Removal (Slide 2 of 4)

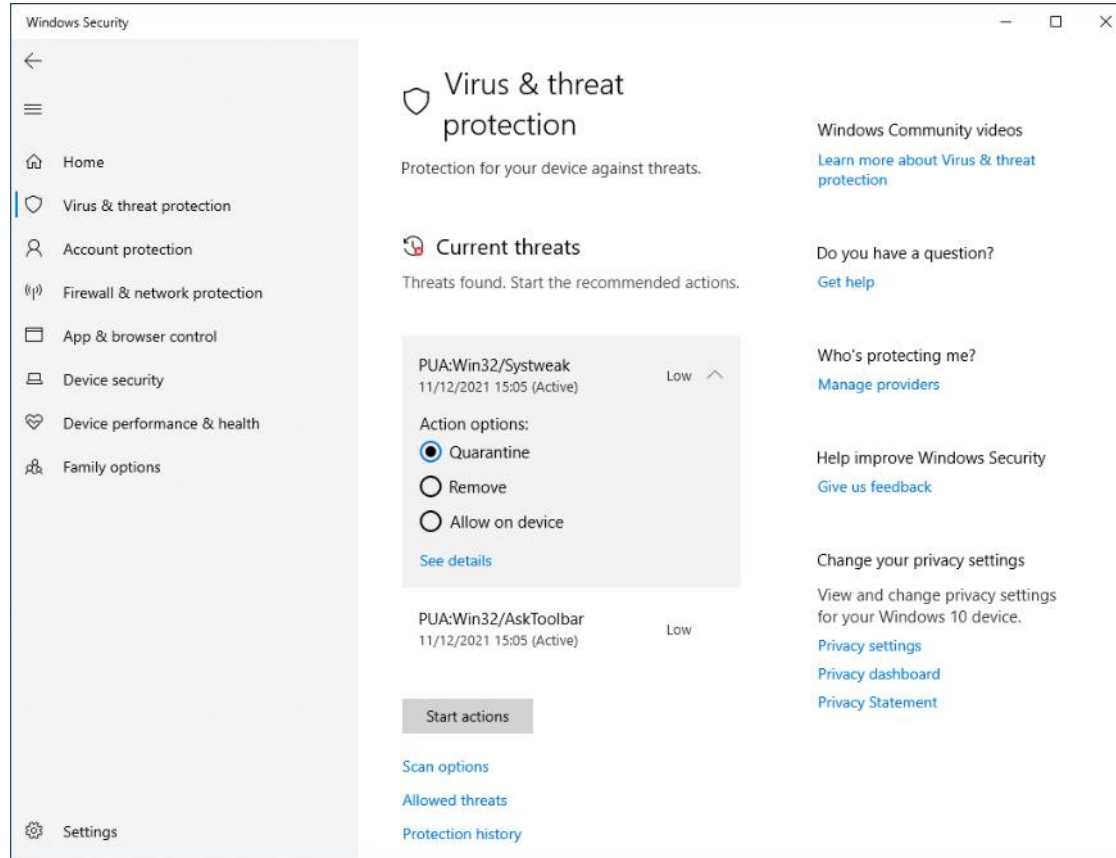
6. Enable System Restore and create a restore point in Windows.

- Newer Windows OS prefer that a user use the File History or Reset Windows options instead of using system restore.

7. Educate the end user.

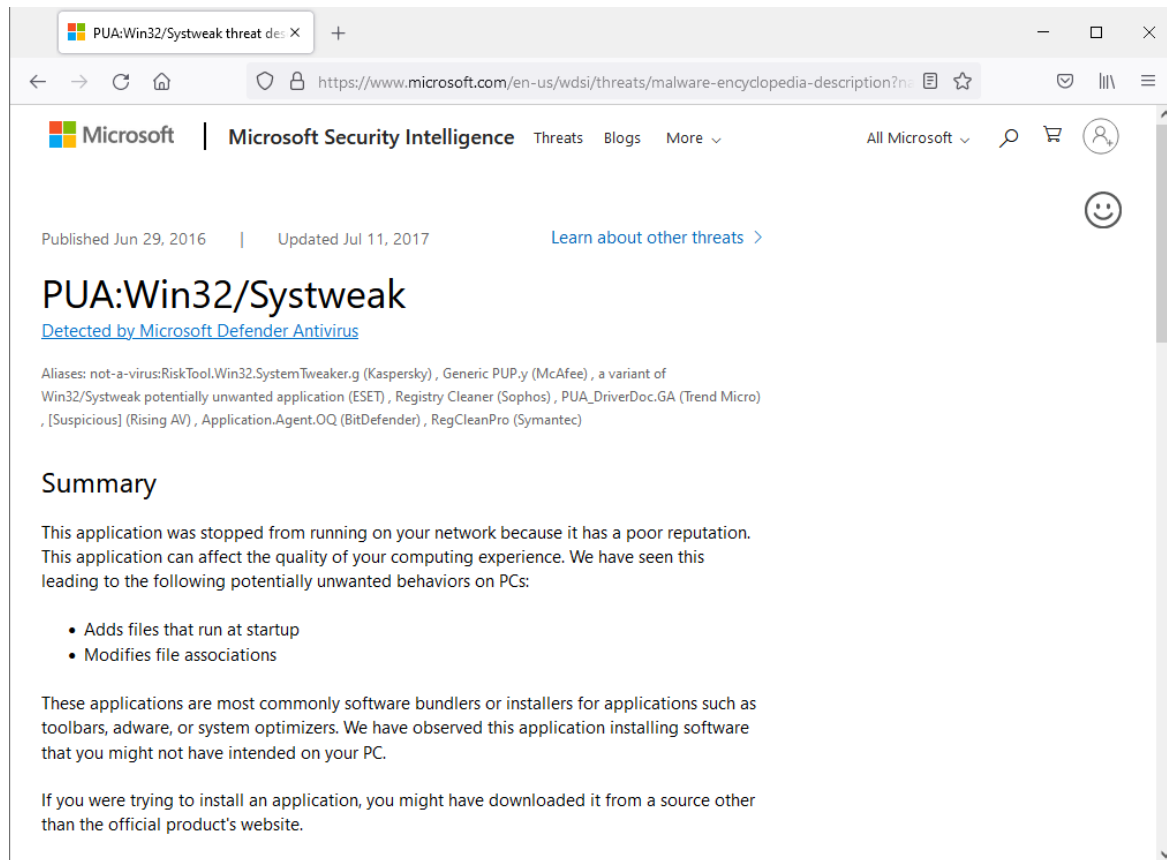
- Most malware is discovered via on-access scanning by an antivirus product. If the malware is sophisticated enough to evade automated detection, the symptoms listed above may lead you to suspect infection.

Best Practices for Malware Removal (Slide 3 of 4)



Screenshot courtesy of Microsoft.

Best Practices for Malware Removal (Slide 4 of 4)



Screenshot courtesy of Microsoft.

Infected Systems Quarantine

Quarantine is not removal

Isolate the system from the network

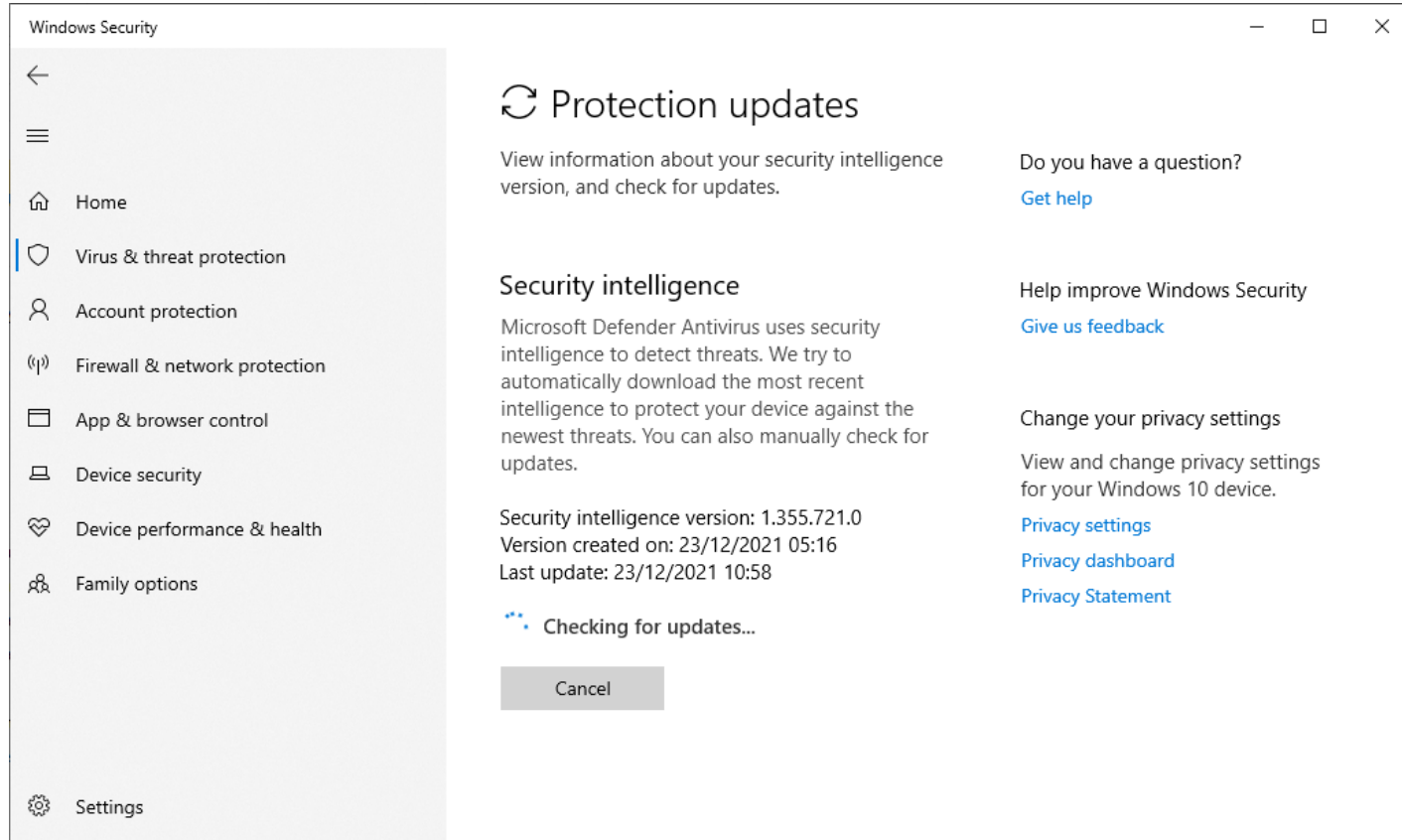
- Physical isolation, if possible, logical isolation at a minimum

Disable System Restore

Malware Removal Tools and Methods (Slide 1 of 2)

- Updated antivirus/anti-malware software
- Use of a third-party application or removal tool
- Scan from another system
- Recovery options
 - WinRE, restore from backup
 - Backups should be carefully evaluated, as they may also contain the malware
- OS reinstallation
 - Erase, repartition, reinstall

Malware Removal Tools and Methods (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Malware Infection Prevention

On-access scanning

Schedule regular scans

Re-enable System Restore and Services

Educate end user

- Do not assign blame
- Educate not alienate

Lab Activity

- Challenge Lab: Resolve Security Tickets
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Workstation Security (Slide 1 of 2)

- Malware Vectors
- Malware Payloads
- Ransomware and Cryptominers
- Troubleshoot PC Security Symptoms
- Desktop Alerts and Notifications
- Endpoint Monitoring Solutions
- Troubleshoot Browser Symptoms
- Best Practices for Malware Removal
- Infected Systems Quarantine

Lesson Review: Workstation Security (Slide 2 of 2)

- Malware Removal Tools and Methods
- Malware Infection Prevention